

Python 4-Week Mastery Roadmap

Revision-focused, practice-heavy plan — built for someone who knows the basics but needs to rebuild fluency before moving into **Data Science** & **Machine Learning**

Week 1: Core Fundamentals — Revise + Drill

Goal: **Variables, data types, operators, conditionals, loops** — at muscle-memory level

Day 1-2	Variables, data types , type casting, operators → 15-20 small practice problems (calculator, temperature converter, etc.)
Day 3-4	Conditionals + Loops → logic-heavy problems (FizzBuzz, prime number checker, pattern printing, number guessing game)
Day 5-6	Functions (params, return, default args, *args/**kwargs) → 10 function-based problems
Day 7	Mini project — combine everything from Day 1-6 (e.g. simple quiz game or unit converter tool)

Week 2: Data Structures — Deep Practice

Goal: **Lists, tuples, dicts, sets** — practiced enough to use without thinking

Day 1-2	Lists + list comprehensions → sorting, searching, manipulation problems
Day 3-4	Dictionaries + Sets → word frequency counter, duplicate remover, grouping problems
Day 5-6	String manipulation (very important for real projects) → palindrome, anagram, text processing problems
Day 7	Mini project — e.g. contact book or inventory management (using list + dict)

Week 3: OOP + File Handling + Error Handling

Goal: Understand real-world code structure

Day 1-3	OOP (classes, objects, inheritance, polymorphism, encapsulation) → practice by building small classes: banking system, library management
Day 4	File handling (read/write CSV, txt files)

Day 5	Exception handling (try/except/finally) — handling real errors
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Day 6-7	Mini project — combine everything (e.g. student record system with file storage + OOP + error handling)
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Week 4: Advanced + Data Science Prep

Goal: Bring Python to “production-ready” level and smoothly transition into Data Science

Day 1-2	Lambda, map/filter/reduce , advanced list/dict comprehensions
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Day 3-4	Modules, packages , virtual environments, pip
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Day 5-7	NumPy intro (array basics) — to build the foundation for the next phase (Data Science)
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Key Principle

A forgotten concept, once re-learned a third time, tends to stick permanently. Use this cycle for every concept:

1. **Revise** the concept (5–10 minutes)
2. Immediately **solve** 3–5 practice problems
3. Next day, **re-solve** one problem from the previous day (spaced repetition)